



Introduction to Git and GitHub

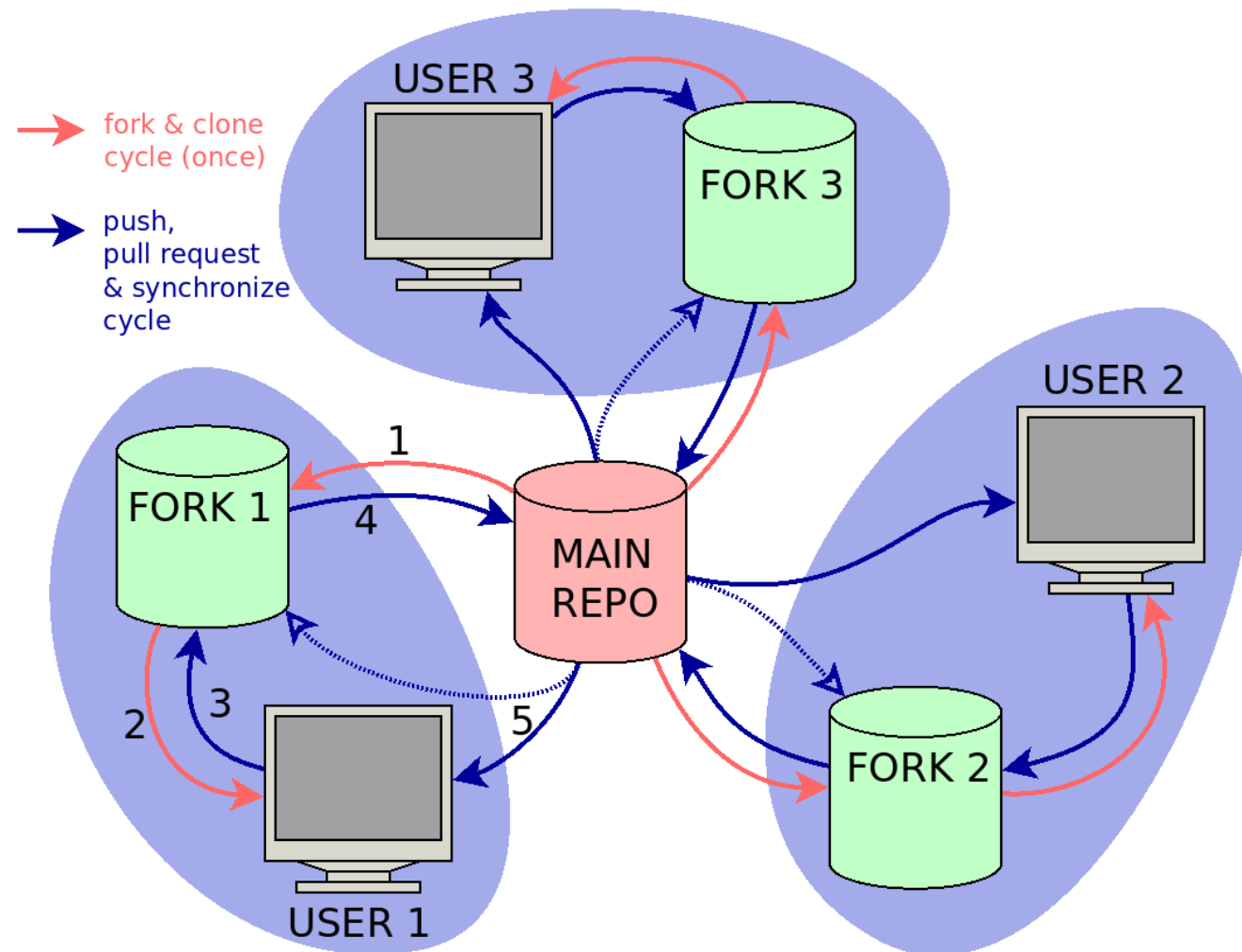
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What is Forking?

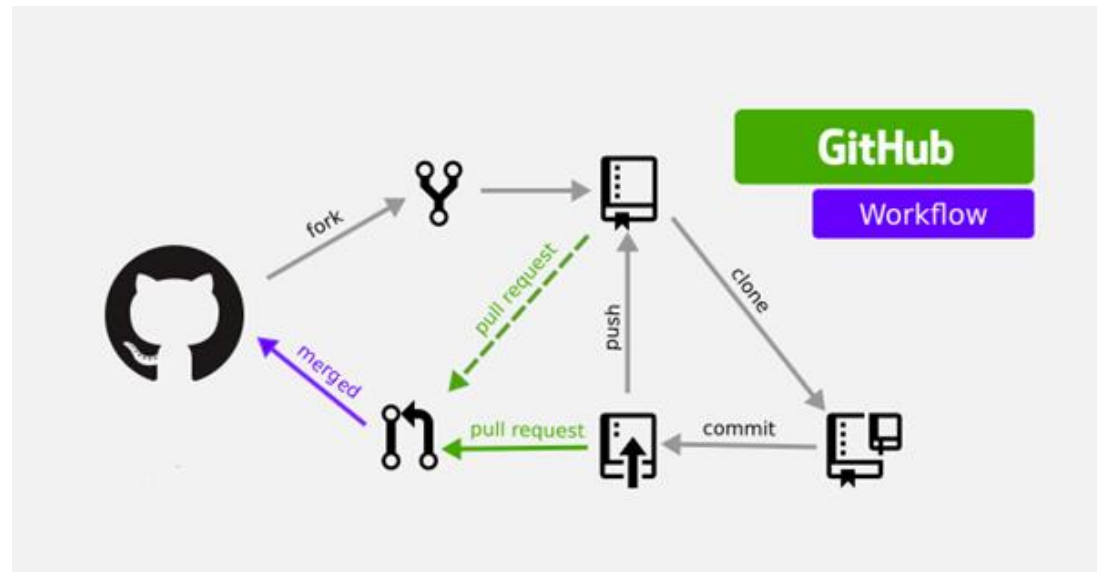


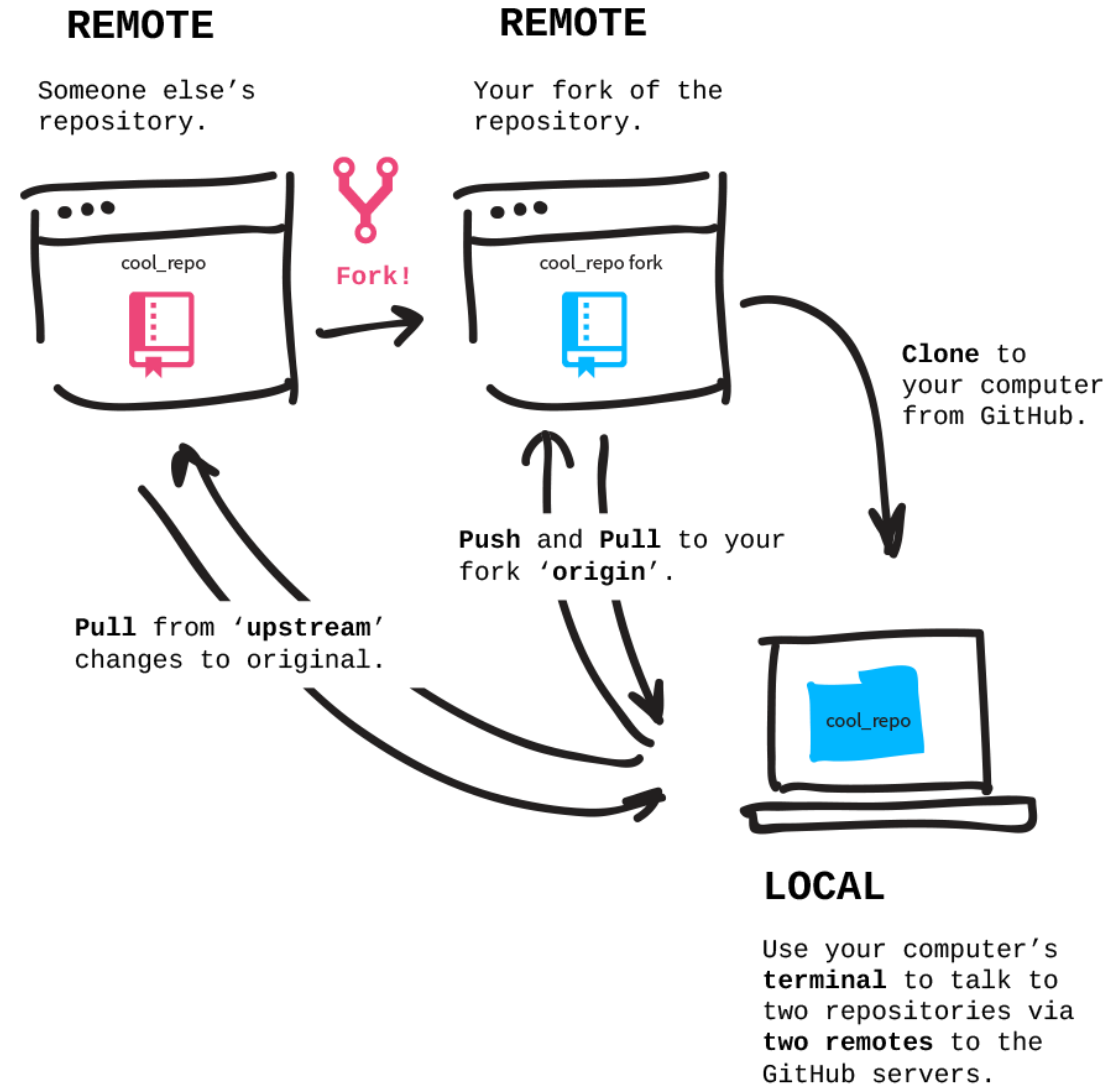
Forking



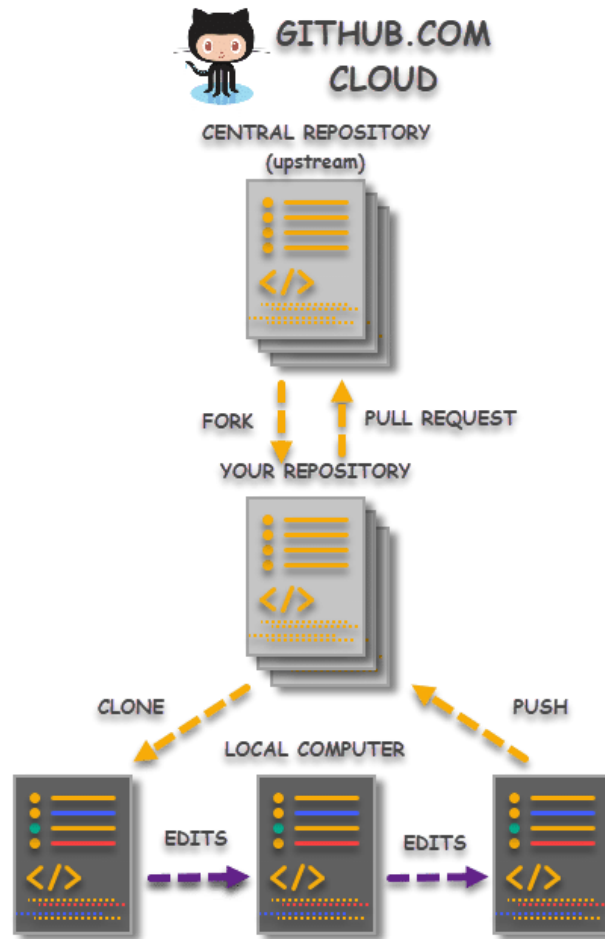
Forking

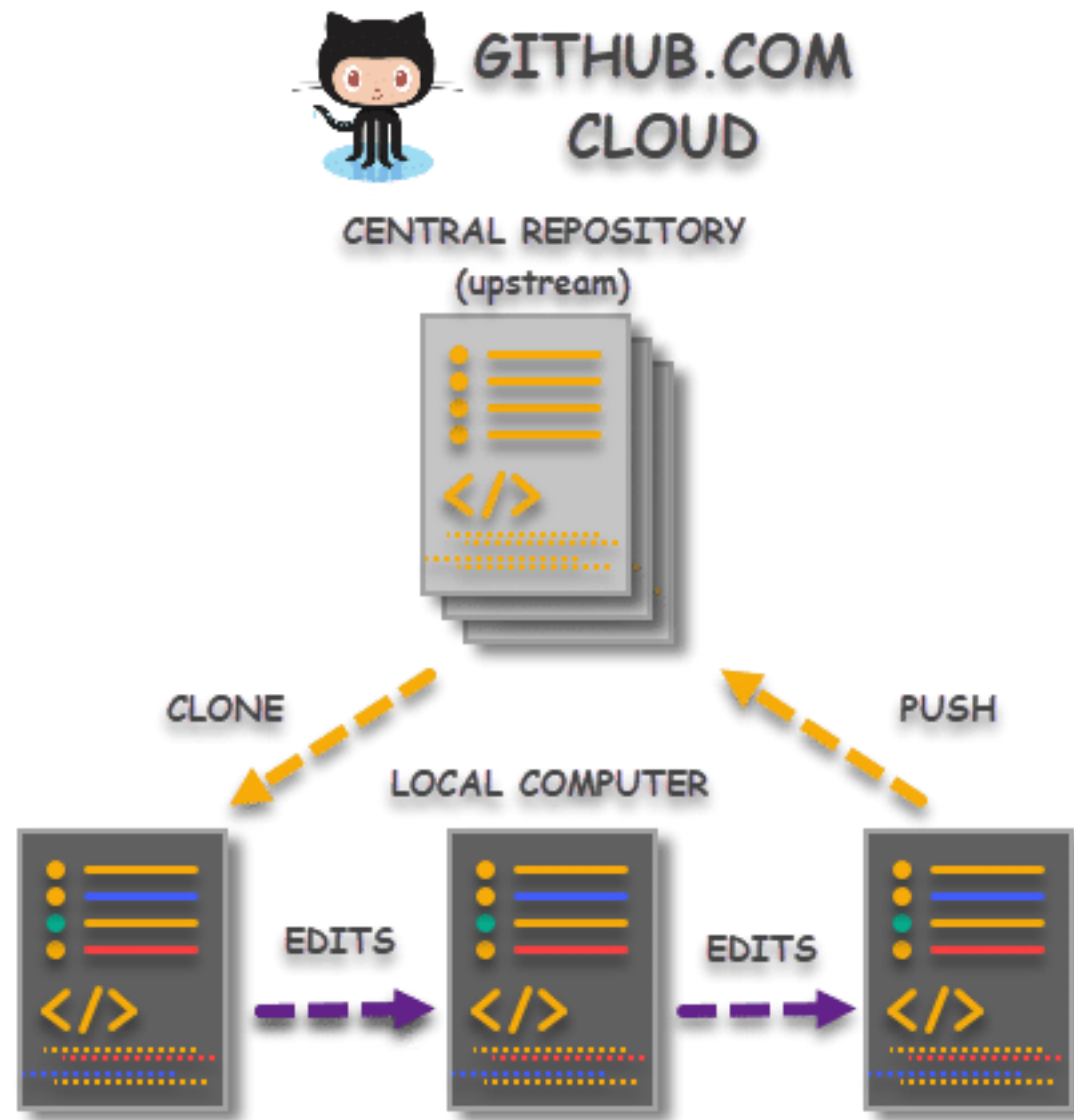
- Create a personal copy of a repository on GitHub without affecting the original project.
- The forked repository is "downstream" from the original "upstream" repository.
- Make changes freely on your fork without altering the original project.
- Suggest changes to the original repo via a pull request.
- Fork a repo, fix bugs, and submit a PR to request merging into the upstream repository.
- Keep your fork up to date by pulling changes from the upstream repo





Forking





Fork Vs Clone

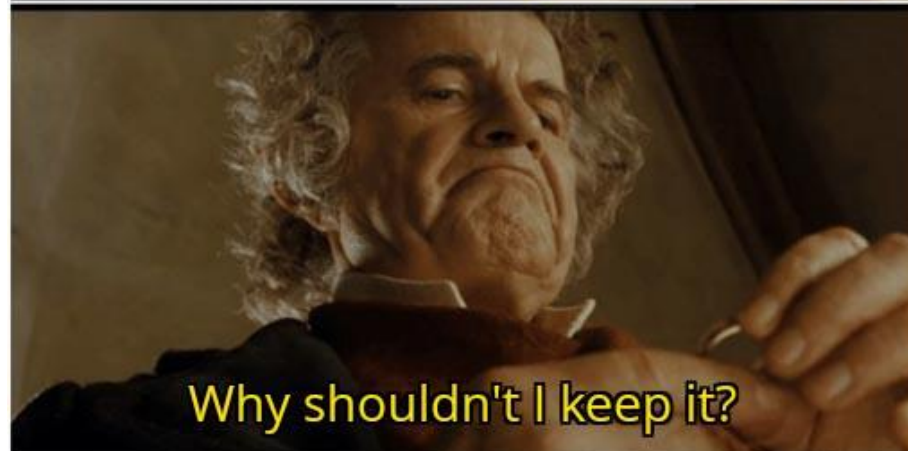
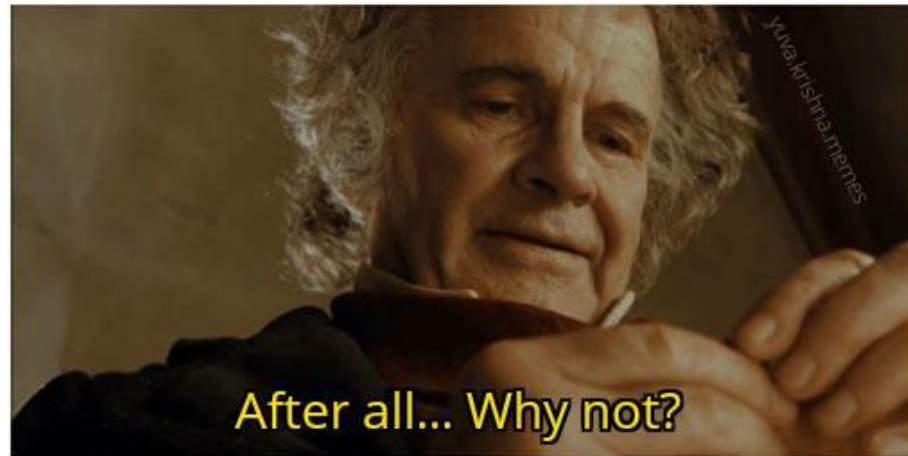


Fork Vs. Clone

“Think of cloning as borrowing a repo. Forking is like copying it and making it yours”



Me after forking all major
Opensource projects into my GitHub
account without any reason..



Real World Use Cases

- Rolling Back a Buggy Release
 - git revert
 - git reset
 - git log
- Experimenting with new features
 - Branching
 - Merging
- Maintaining a Clean Commit History
 - git rebase -I
 - git commit --amend

git revert <commit_hash>

**git reset --<mode>
<commit_hash>**

**Mode could be soft, mixed
and hard**

Best Practices & Tips

- Always follow your organization's policies and avoid pushing potentially confidential information to GitHub.
 - In general, you should only track code on GitHub.
 - Consider whether a repository should be private or public.
- Use README files to document your repositories.
- Write descriptive commit messages to help your collaborators and future self.
- Work with your collaborators to develop workflows to meet your needs.
- Learn about other useful features from GitHub (e.g., [manage projects](#), [host websites](#), [automate tasks](#), and much more).

Next Steps

- These are some of our favorite resources to learn more:
 - [GitHub's Getting Started Guide](#)
 - [Happy Git and GitHub for the UseR](#)
 - [Getting Git Right](#)
- This is a great interactive exercise to practice what you learned today:
 - [GitHub Skills – Introduction to GitHub](#)